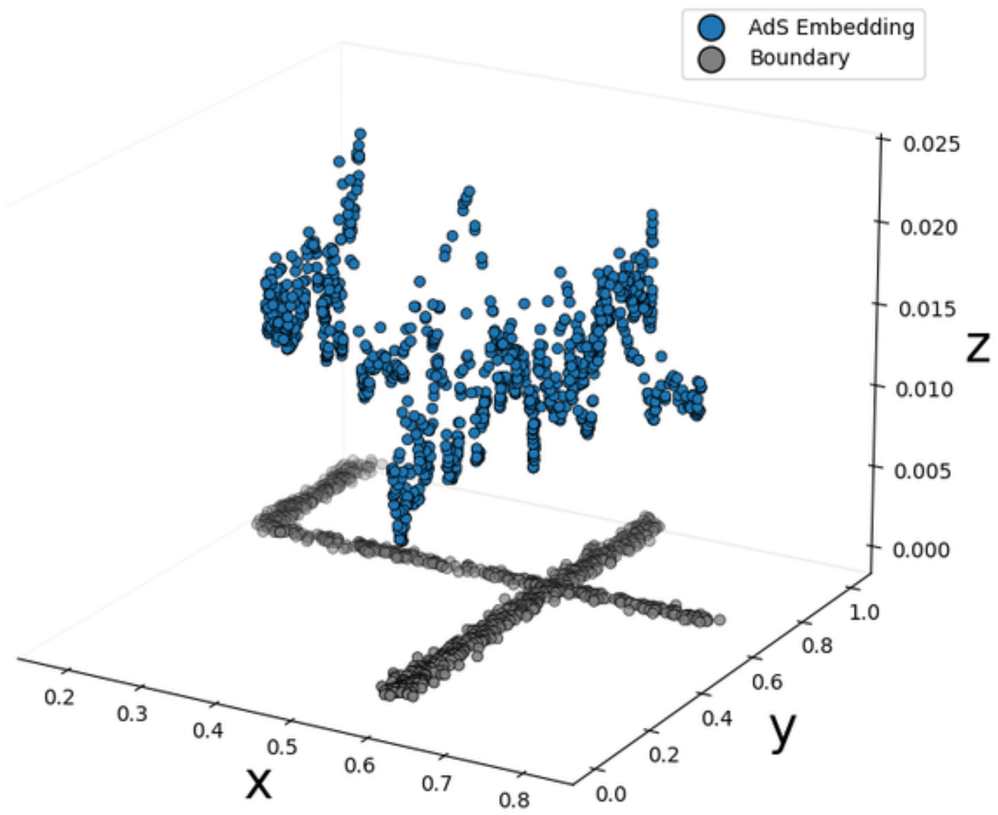
AdS-GNN Step 1: Details on Lifting

$$ds^2 = \frac{1}{z^2}(dz^2 + dx^2 + dy^2)$$

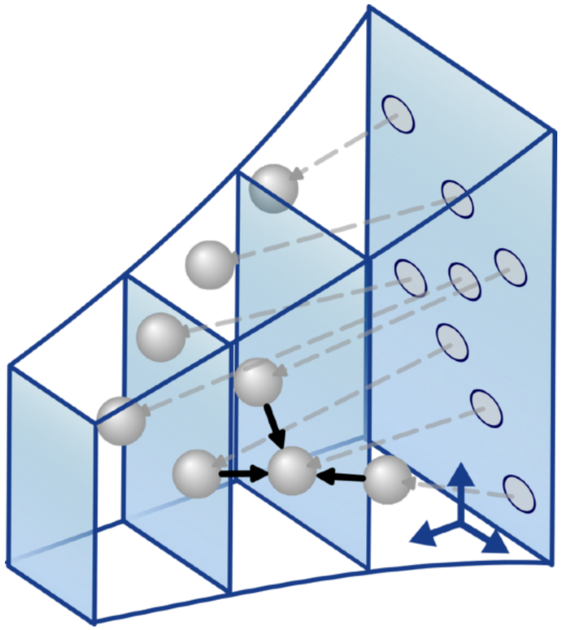


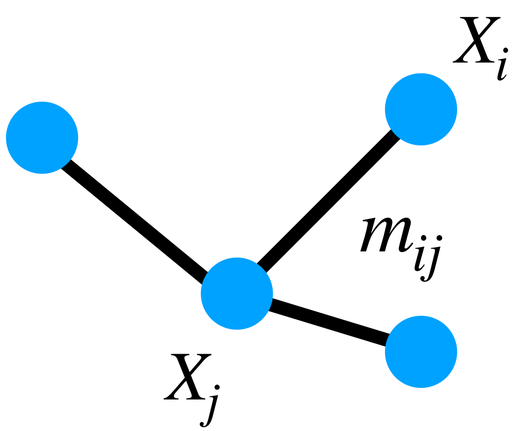
To embed a point into AdS, we need to pick a cutoff $z_0 \neq 0$.

1. Embed each point with a small finite $z_i = z_0$.
2. Find the k nearest neighbours of each point.
3. Compute their AdS "center of mass" (Galperin 1993) to get a $(\bar{x}_i, \bar{z}_i)$.
4. Embed the point as $X_i = (x_i, \bar{z}_i)$.

Physically: this sets the "length scale" of each point to be the average distance to its neighbours. (Does not depend much on initial z_0).

AdS GNN Step 2: Details on bulk operations





We now operate on the points in the bulk using standard **graph neural network** ideas:

1. Each **node** at position X_i has scalar degrees of freedom h_i and is connected to k nearest neighbours by **edges**.

2. Each **edge** carries messages m_{ij} derived from the data on the nodes that they connect.

3. Layers work with **message-passing** as follows:

$$m_{ij}^{(l)} = \psi_e^{(l)}(h_i^{(l)}, h_j^{(l)}, D(X_i, X_j))$$
$$h_i^{(l+1)} = \psi_h^{(l)}(h_i^{(l)}, m_{ij}^{(l)} \,|\, i \in \mathcal{N}(j))$$

where **D is AdS-invariant distance**. At the end, take an average (or other permutation-invariant operation) over all nodes.